

★ WEEK 6 · PART 1 of 3

The VAR Model

When Variables Move Together

Vector autoregression · the coefficient matrix · stability · joint forecasting

Univariate models treat each series in isolation, but markets are interconnected. Vector autoregression lets several series interact — each one's future shaped by every series' past.

1

PART 1 · THE VAR MODEL

ARMA, for a Vector of Series

Why we model series jointly, the VAR(p) equation and its coefficient matrix, the stability condition, lag selection, and forecasting several variables at once.

The Man Who Told Economists to Stop Pretending — Part 1

A 1980 paper that overturned macro modeling

Christopher Sims argued that the huge structural macro models of the day rested on 'incredible' assumptions — theory dressed up as fact. He proposed letting the data speak instead.

The problem he saw

- Big models imposed hundreds of untested theoretical restrictions.
- Those restrictions were assumed, not evidenced.
- Forecasts and policy advice inherited every hidden assumption.

His alternative

- Treat all variables symmetrically — none privileged by theory.
- Let each depend on the recent past of them all.
- That minimal-assumption model is the VAR.

The Man Who Told Economists to Stop Pretending — Part 2

The VAR revolution, and the lesson

Sims' VAR became the workhorse of empirical macro and finance, and earned him the 2011 Nobel. Its lesson: model the relationships you can measure, not the ones you assume.

What it changed

- VAR became the standard tool for studying co-movement.
- Impulse responses and causality tests grew out of it.
- It reshaped how economists read data — humbly.

The lesson for us

- Let several series interact, with minimal theory imposed.
- Measure the dynamics; don't assume them.
- But stay humble about what 'causality' really means (Part 2).

Why Go Multivariate

Three reasons one-at-a-time is not enough

1

Markets are interconnected

Rates move stocks; oil moves airlines; a shock to one asset ripples into others. A model of one series alone is blind to all of it.

2

Univariate models miss the links

Even our GARCH work (Week 5) modeled each series' risk in isolation. Portfolios and pairs depend on how series move together.

3

VAR captures the feedback

By letting every variable depend on every variable's past, a VAR models lead-lag, feedback, and co-movement in one system.

Part 1 in one line: stop modeling one series alone — let several interact in a single system.

Why Model Series Together?

Isolated models miss the connections that matter

When variables influence each other, modeling them separately throws away information. A joint model uses one series' past to help forecast another — and to study how shocks spread.

What you gain

- Better forecasts, by borrowing strength across series.
- A way to test who leads whom (Granger causality, Part 2).
- A map of how a shock propagates through the system (IRF, Part 3).

Everyday examples

- Short and long interest rates that move together.
- A stock and its sector index.
- Spot and futures prices — the seed of a pairs trade.

The VAR(p) Model

ARMA, but for a vector of variables

$$\mathbf{Y}_t = \mathbf{c} + A_1 \mathbf{Y}_{t-1} + A_2 \mathbf{Y}_{t-2} + \cdots + A_p \mathbf{Y}_{t-p} + \boldsymbol{\varepsilon}_t \quad \text{VAR}(p)$$

Reading it

- $\mathbf{Y}_t$ is a VECTOR — several series stacked together.
- Each A is a MATRIX of coefficients, one per lag.
- Every variable regresses on the past of ALL variables.

Why it generalizes AR

- With one variable it collapses to an ordinary AR(p).
- The matrices A hold both own-effects and cross-effects.
- p is the lag order, exactly as before.

A Bivariate VAR(1), Written Out

Two equations make the idea plain

$$\begin{aligned}y_{1,t} &= c_1 + a_{11}y_{1,t-1} + a_{12}y_{2,t-1} + \varepsilon_{1,t} \\y_{2,t} &= c_2 + a_{21}y_{1,t-1} + a_{22}y_{2,t-1} + \varepsilon_{2,t}\end{aligned}$$

The two equations

- Each series depends on its OWN last value (a_{11} , a_{22}).
- And on the OTHER series' last value (a_{12} , a_{21}).
- Two coupled equations, estimated together.

Where the magic is

- $a_{12} \neq 0$ means Y2's past helps predict Y1.
- $a_{21} \neq 0$ means Y1's past helps predict Y2.
- Those cross-terms are the whole point of a VAR.

Reading the Coefficient Matrix

Where the cross-dynamics live

$$A_1 = \begin{pmatrix} a_{11} & \overbrace{a_{12}}^{\text{CROSS}} \\ \underbrace{a_{21}}_{\text{CROSS}} & a_{22} \end{pmatrix} \quad \text{off-diagonals carry the interactions}$$

Diagonal vs off-diagonal

- Diagonal terms are the usual own-lag (AR) effects.
- Off-diagonal terms carry the interactions between series.
- A diagonal matrix means NO interaction — separate ARs.

How to read it

- A large off-diagonal entry = a strong lead-lag link.
- Its sign tells you the direction of the influence.
- The whole matrix is a compact map of the system.

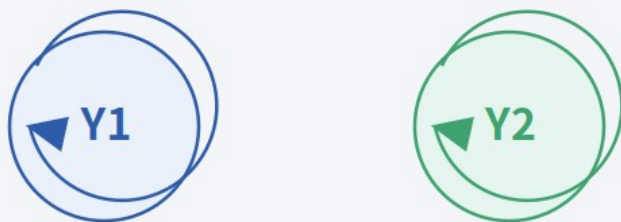
Univariate vs VAR: Who Listens to Whom

Separate models let each series hear only itself; a VAR lets every series hear every series' past — so feedback and co-movement are finally captured.

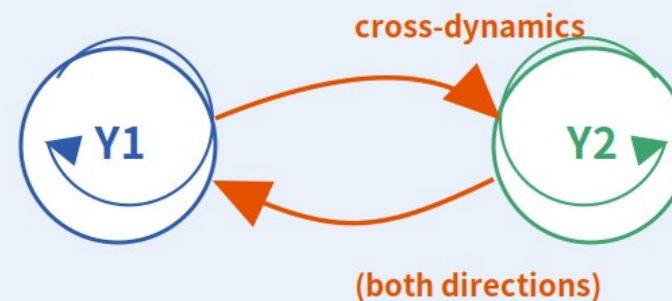
Univariate vs VAR: Who Listens to Whom

In a VAR, every variable's forecast uses every variable's past - the links univariate models ignore.

Two univariate models
each series listens only to ITSELF



One VAR
each series listens to BOTH series' past



feedback and co-movement are captured

What a VAR Captures

Feedback, lead-lag, and co-movement

The dynamics it sees

- Feedback loops: X moves Y, which later moves X back.
- Lead-lag: one series systematically anticipates another.
- Common responses to shared, underlying forces.

Why it matters in finance

- Risk and return spill across related assets.
- Hedges and pairs depend on these joint dynamics.
- A single-series model literally cannot represent them.

The Reduced Form: Let the Data Speak

Minimal theory, maximal honesty

A reduced-form VAR imposes almost no economic theory — just that variables depend on their recent joint past. It measures relationships rather than assuming them.

What 'reduced form' means

- No variable is singled out as the cause of another.
- The equations describe correlations through time, not mechanisms.
- It is deliberately agnostic about the 'true' structure.

The trade-off

- You get honest, data-driven dynamics and forecasts.
- But you cannot read causal 'shocks' off it directly.
- Recovering those needs extra assumptions (Part 3).

Estimating a VAR

Surprisingly simple: one equation at a time

The method

- Each equation has the SAME regressors — all lagged variables.
- So you can estimate each one by ordinary least squares.
- Equation-by-equation OLS is efficient here.

What you get

- The coefficient matrices A and the residual covariance.
- That residual covariance matters later (correlated shocks, Part 3).
- Software fits the whole system in one call.

Stability of a VAR

The multivariate version of the unit-root condition

stable $\iff$ all eigenvalues of the companion matrix lie INSIDE the unit circle

The condition

- Stack the VAR into a big 'companion' matrix.
- All its eigenvalues must lie inside the unit circle.
- For one variable this reduces to $|\phi| < 1$.

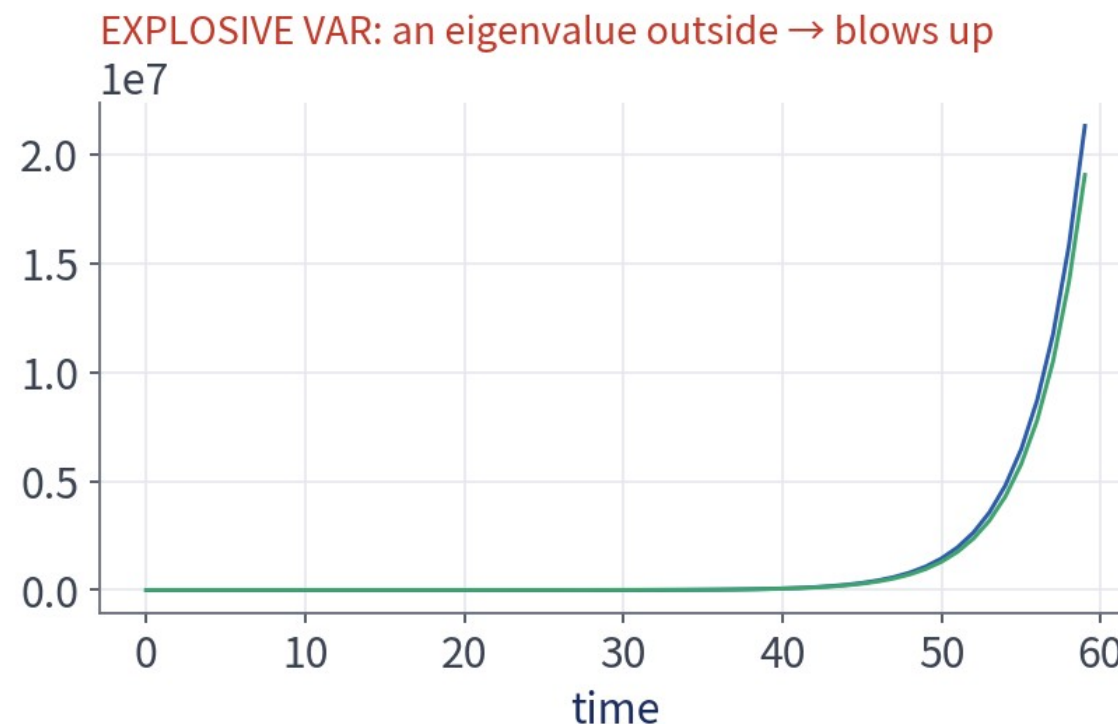
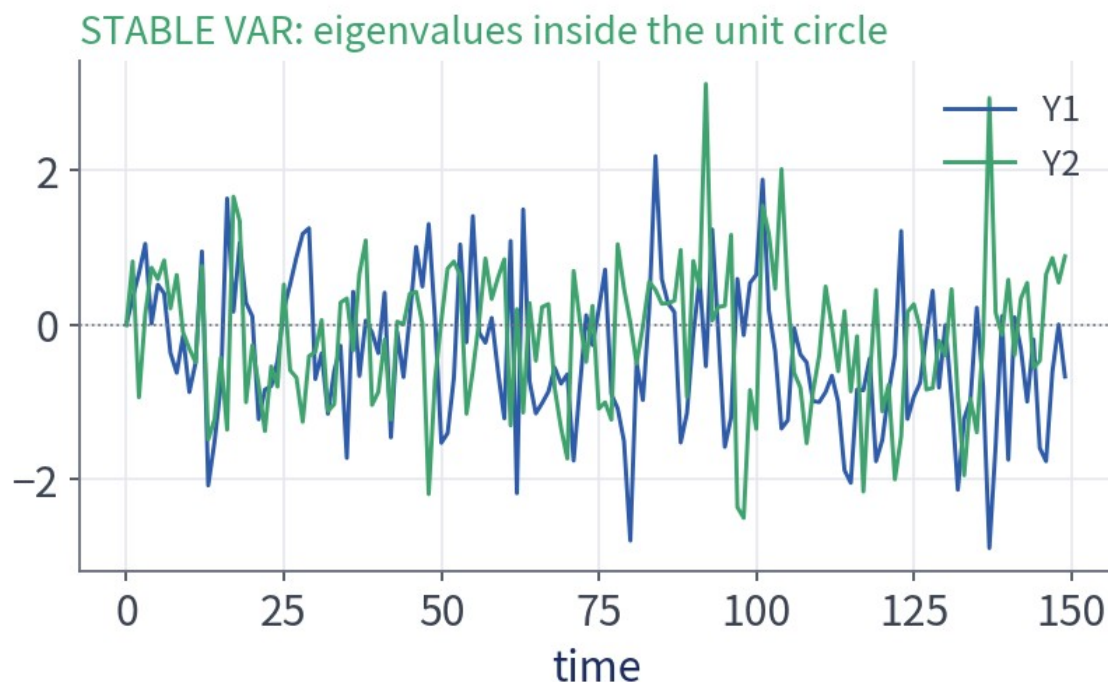
Why it matters

- A stable VAR is stationary — its shocks eventually fade.
- An unstable VAR explodes, exactly like a unit root.
- Check stability before trusting any forecast or IRF.

Stable vs Explosive

With all eigenvalues inside the unit circle (left), the system fluctuates around a fixed level. Push one outside (right) and the whole system blows up.

Stability is the multivariate unit-root condition: all eigenvalues must lie inside the circle



Stationarity Still Comes First

Apply a VAR to stationary series

Like ARMA, a standard VAR assumes stationary inputs. If the series trend, difference them first — or, if they move together, a special structure awaits (Week 7).

The usual prep

- Test each series for a unit root (Week 2).
- Difference trending series to returns or growth rates.
- Then fit the VAR on the stationary transforms.

A tantalizing exception

- If two $I(1)$ series are cointegrated, differencing loses information.
- Their long-run link deserves its own model.
- That is the vector error-correction model — Week 7.

Choosing the Lag Order p

Balancing fit against an explosion of parameters

choose $p = \arg \min \text{ AIC / BIC / HQ; } \quad \# \text{params} = k + p k^2$

How to choose

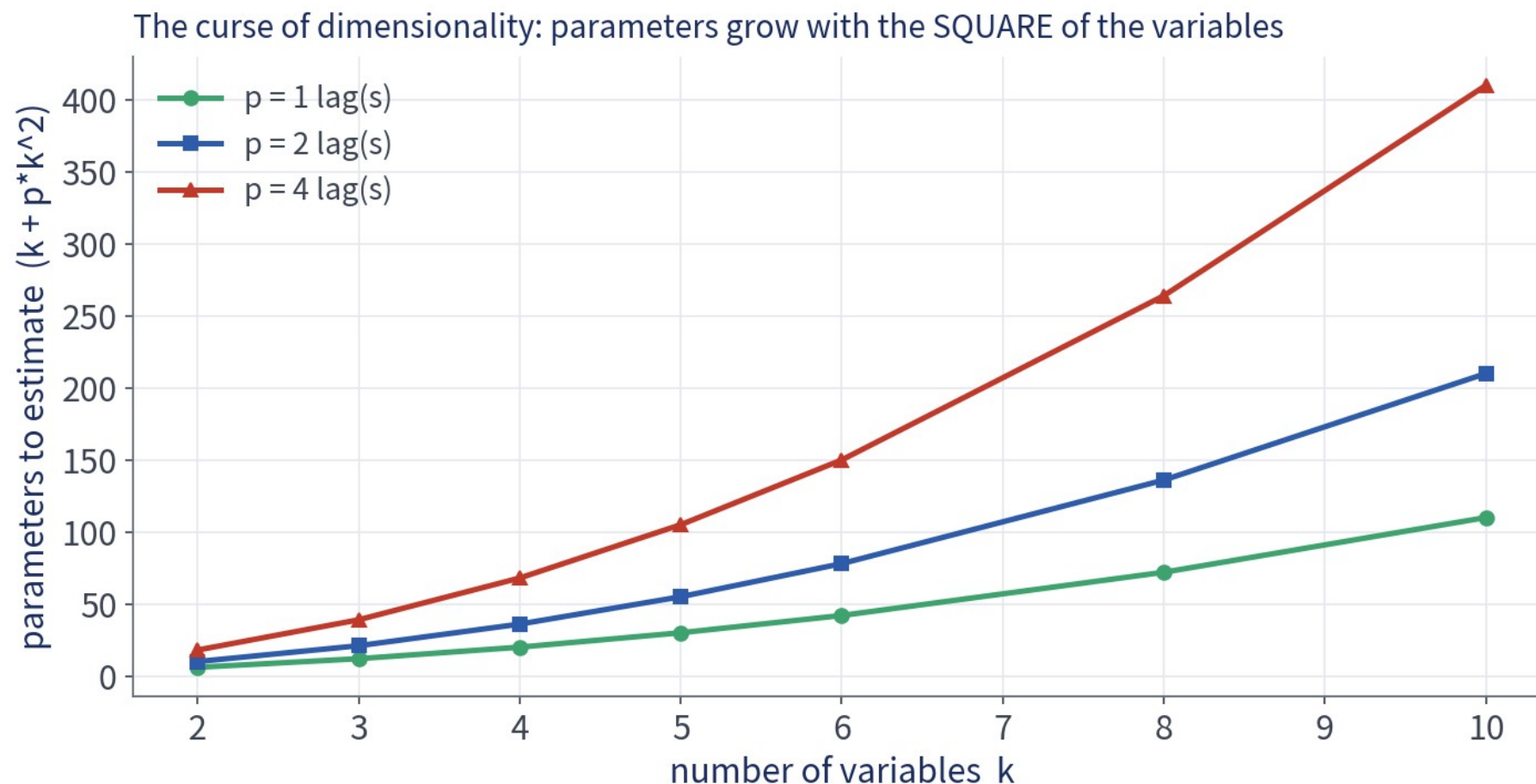
- Fit several lag orders and compare AIC, BIC, or HQ.
- Lower is better; BIC leans toward fewer lags.
- Confirm the residuals show no leftover autocorrelation.

The dimensionality trap

- Parameters grow with the SQUARE of the variable count.
- A few variables and a few lags is already a lot to estimate.
- Parsimony matters even more than in the univariate case.

The Curse of Dimensionality

The parameter count grows like $k^2 \cdot p$. Doubling the number of variables roughly quadruples the coefficients — which is why VARs are kept small and low-order.



Forecasting With a VAR

Many variables, forecast jointly

$$\hat{\mathbf{Y}}_{t+1} = \mathbf{c} + A_1 \mathbf{Y}_t + \cdots + A_p \mathbf{Y}_{t-p+1} \quad (\text{iterate for } h > 1)$$

How it works

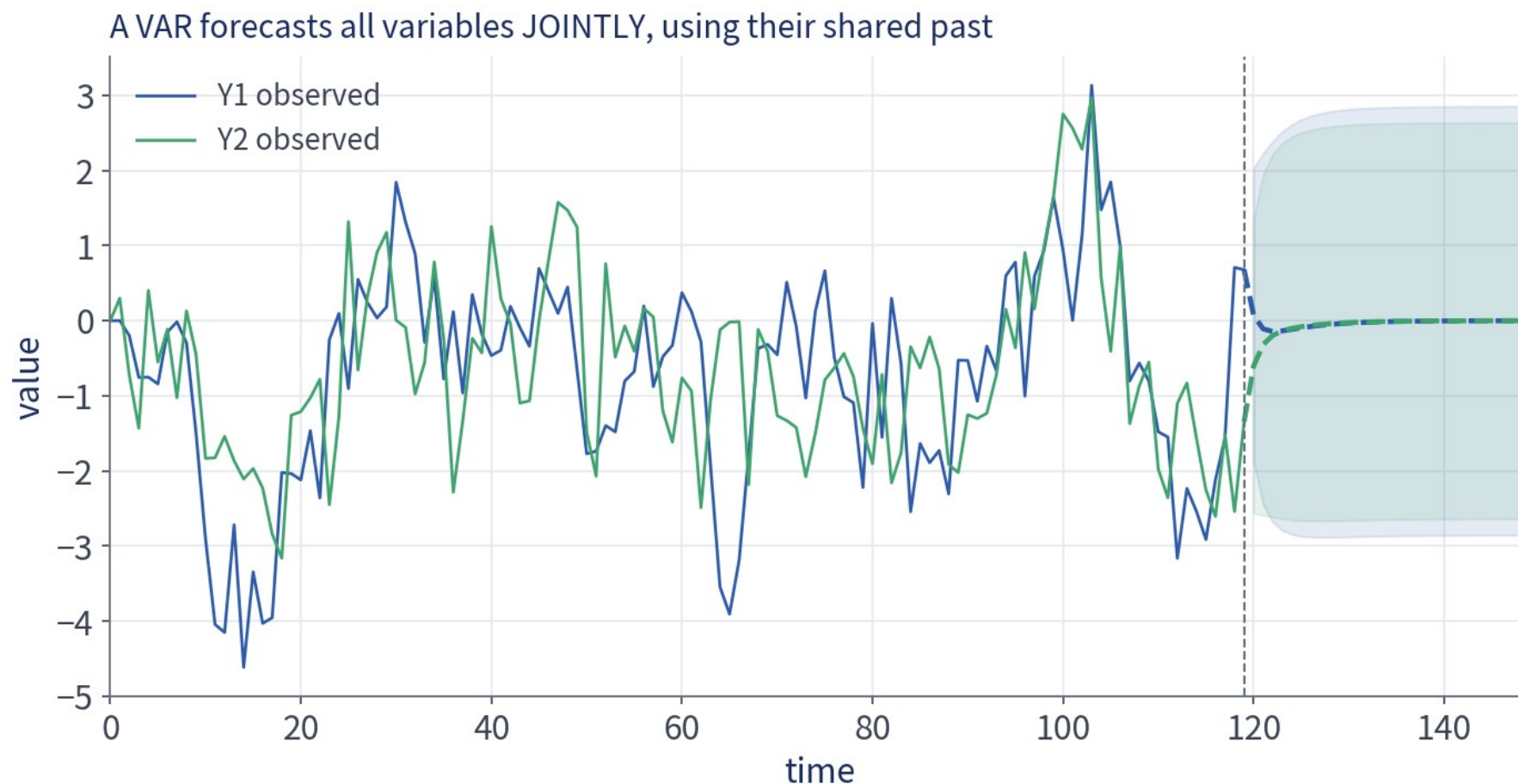
- One step ahead uses the last observed values of all series.
- Longer horizons feed the forecasts back in, iterating.
- Every series is forecast at once, consistently.

Why joint helps

- Each forecast exploits the others' recent moves.
- The forecasts respect the cross-dynamics you estimated.
- Intervals widen with horizon, as always — but jointly.

A VAR Forecasts Everything at Once

Both series are projected jointly, each using the other's recent past. The intervals fan out with the horizon, reflecting the accumulating shocks.



The VAR Workflow

Five steps from several series to a fitted system

1 · Stationarize

each series

Test and difference so every series is stationary.

2 · Select p

lag order

Compare AIC/BIC/HQ; keep p small.

3 · Estimate

OLS system

Fit equation-by-equation; get A and the residuals.

4 · Check

stability

Confirm eigenvalues inside the circle; clean residuals.

5 · Use

forecast+analyze

Forecast jointly; then Granger, IRF, FEVD.

Steps 1–4 build the model; step 5 — the interesting part — is Parts 2 and 3.

VAR vs Separate Univariate Models

What the joint model buys, and costs

Aspect	Separate models	VAR (joint)
Cross-dynamics	ignored	captured via off-diagonal terms
Forecasts	each series alone	all series jointly and consistently
Lead-lag / feedback	invisible	explicit and testable
Parameter cost	few	many — grows with k squared

VAR buys realism about interactions at the price of far more parameters — so keep k and p small.

Fit a VAR and Forecast

Direct Claude Code from several series to a joint model

GOAL Build a VAR on two or three related series, choose the lag order, check stability, and forecast them jointly.

[INPUT] Prompt · Claude Code

Write and run `var_lab.py` that:

1. downloads 2-3 related series (e.g. a stock, its sector ETF, and a rate) and makes each stationary (returns / differences);
 2. selects the VAR lag order p by AIC and BIC with `statsmodels`, and fits the VAR;
 3. checks stability (eigenvalues) and residual autocorrelation;
 4. forecasts all series 20 steps ahead with joint confidence bands, and plots them.
- Report the chosen p and whether it is stable.

How to work with it

- Confirm every series is stationary first.
- Watch how AIC and BIC can disagree on p .
- All eigenvalues inside the circle = stable.
- Keep it; Part 2 adds Granger causality.

Deliverable: `var_lab.py` + a joint forecast of your chosen series.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

In a bivariate VAR, what does a non-zero off-diagonal coefficient mean?

One series predicts the other

Q2

How do you check whether a fitted VAR is stable?

Eigenvalues inside circle

Q3

Why do VARs force you to keep the number of variables small?

Params grow with k^2

All three come from the coefficient matrix, stability, and the dimensionality cost.

Three Misconceptions to Avoid

Where VAR intuition slips

Common myths

- "More variables always make a VAR better."
- "A VAR reveals what causes what."
- "You can fit a VAR straight to price levels."

The reality

- Each added variable multiplies the parameters — VARs stay small.
- A reduced-form VAR shows prediction, not causation (Part 2).
- Standard VARs need stationary inputs — difference first (or Week 7).

Three Things to Carry Forward

Part 1 in one screen

1

A VAR is ARMA for a vector

Every series depends on the recent past of all series. The off-diagonal coefficients carry the cross-dynamics that univariate models throw away.

2

Stability and parsimony are the constraints

All eigenvalues must sit inside the unit circle, and the parameter count grows with k^2 — so keep the system small and stationary.

3

The joint model unlocks new questions

With a fitted VAR we can forecast jointly and then ask who leads whom and how shocks propagate — the subjects of Parts 2 and 3.

Next — Part 2: Granger causality, testing whether one series' past truly helps predict another.